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Sampling is the process of selecting some elements in a larger population. A sample is a smaller group which is a representation of the population. The purpose is to gather data from the sample that can undergo analysis to produce meaningful conclusions about the population.

Sampling a key step in Survey research designs. Other types of research that require data collection from participants or other sources utilize sampling.

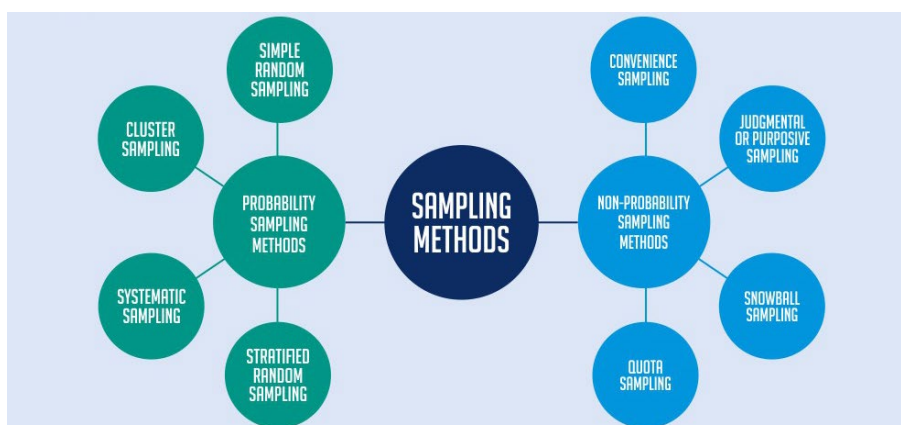
1. The need for sampling

Myers (n.d) suggests reasons for sampling in research:

- a. High cost of using the entire population
- b. Long time frame to collect data might render IT research obsolete by then the results become available
- c. The exact number of the population might not be known hence, probabilities and statics may be flawed

When the population is small then there is no need for sampling.

There are two major types of sampling techniques, probability sampling and non-probability sampling. The difference between the two is the ability for *randomization*.



Different sampling techniques. Source: QuestionPro available at <https://www.questionpro.com/blog/>

2. Probability sampling

Also known as Random sampling. This is the most precise type of sampling. The premise is that every member of the population has an equal chance of being selected. The sample elements must have the same characteristics to those elements in the population. Each characteristic can be identified by the researcher allowing him/her to generalize from the sample to the population.

2.1 Simple random sampling

Used when the population is small or when there are no known characteristics of the population. Every element has an equal chance of being selected.

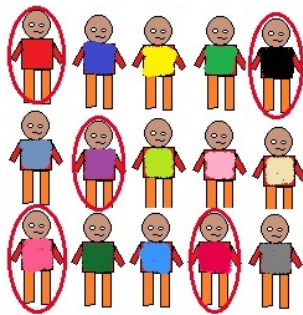


Figure 1: simple random sampling. 5 out of the 25 in the population are randomly selected giving a probability of selection of $1/25$. Source: Singh, N (2018).

2.2 Stratified sampling

The population is split into groups. Elements are then randomly selected from each group. An example might be a population of agile developers but split into groups representing novice, team leaders and dev managers. This will ensure that there is representation from all levels of agile developers.

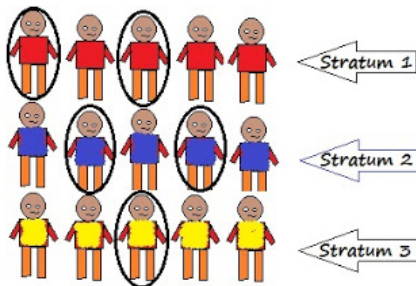


Figure 2: stratified sampling. The population is logically divided into strata. Source: Singh, N (2018).

2.3 Cluster sampling

The population is split into groups. The researcher selects groups at random. Every element from the selected groups form the sample. Cluster sampling is useful when each group is representative of the population. For example, there are 34 academic departments at a university and every academic staff member utilize the same ERP for course management. The researcher may randomly choose 5 from the 34 academic departments but include every academic staff member from that 5.

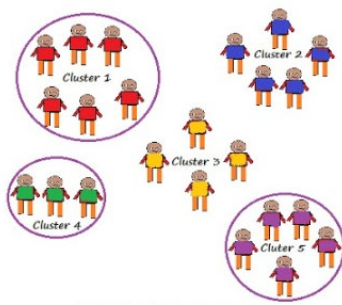


Figure 3: cluster sampling. Source: Singh, N (2018).

2.4 Systematic random sampling

The population is ordered into a list. A starting point is selected at random and then every n^{th} element is chosen for the sample.



Figure 4: systematic random sampling. Source: Singh, N (2018).

Simple algorithm:

1. Assume $N=20$, $n=5$ (N : population size; n : sample size)
2. Divide the population into groups
3. Calculate k (k : number of elements in each group)
 $k = N/n = 4$
4. n_1 = randomly selected first element from the first group (i.e. random start point)
 Assume researcher selects $n_1 = 3$
5. $n_2 = n_1 + k = 3 + 4 = 7$
6. $n_3 = n_2 + k = 7 + 4 = 11$

3. Non-probability sampling

This type of sampling is arbitrary that is, it relies on the researcher's ability to select the sample. There is no element of randomization (random selection) in non-probability sampling. The researcher has to pay careful consideration to potential challenges namely, biased sampling and difficulty in assuring equal opportunity for elements being part of the sample.

3.1 Convenience sampling

The sample is made up of a number of volunteers. Any available element is selected. This type of sampling is sometimes called *accidental* or *haphazard* sampling. It is noteworthy that social sciences and psychology research stress that people who volunteer usually have a similar personality trait that ultimately leads to unreliable results.

It is used when the research does not require any *special* characteristics of the elements. That is, there is no criteria to be part of the sample. The researcher does not consider whether the sample accurately represents the population. An example might be when questionnaires to be handed out to first 50 IT student that the researcher comes across.

3.2 Purposive sampling

The sample is hand-picked based on the aim of the study. For example, should the context of the study be DevOps then only those companies that have implemented this will be included in the sample. In other words, the sample is chosen only on the basis of the researcher's knowledge and judgment.

Used when the researcher has a very specific profile or set of characteristics for the sample. The group that makes up the sample is usually *head-hunted*. An example might be a sample of software development SMEs that have implemented Scrum within the last 3 years.

3.3 Quota sampling

The samples are selected according to some pre-determined standard (quota). The population is divided into *mutually exclusive* sub-groups. The sub-groups are selected according to known criteria. A simple example is male-female sub-groups.

For example, if literature indicated that Java(48%), C#(34%) and Python(18%) are the dominant programming language skills among young developers, then the selected sample must be representative of this. Three mutually exclusive sub-groups will be established, one sub-group each for Java, C# and Python. Now, assume that the sample already consists of 34% of C# developers. Even though more C# developers may still be volunteering, they will be left out because the quota for C# has been reached in the sample.

3.4 Snowball sampling

This technique is sometimes known as *referral* sampling. It is useful when the research requires very unique characteristics of the sample and there are few available or difficult to reach participants. For example, when the researcher gets the first participant who is well experienced with the Racket programming language, the researcher will ask that participant to recommend other experts.

3.5 Expert sampling

Involves the selection of experts or those that demonstrate profound knowledge in the area of interest. It is similar to purposive sampling but there is one difference explained below,

The researcher might convene an expert panel consisting of persons with acknowledged experience and insight into that field or topic and ask them to examine the aspects of the research and comment on their appropriateness and validity. The advantage of doing this is that the researcher is not out on their own trying to defend decisions – they have some acknowledged experts to back them up (Trochim, 2020).

The key difference between expert and purposive sampling is that expert participants through their knowledge, experience and expertise may alter/inform the research.

Tutorial

1. What is the major difference between probability and non-probability sampling?
2. Differentiate between sample, sample frame and population
3. List 2 difficulties of pure random sampling?
4. Provide an example for when each of the following sampling techniques will be used:
 - a. Expert
 - b. Snowball
 - c. Stratified
5. Consider systematic random sampling. Assume that $N=60$, $n=15$. Determine number of elements in each group, k . Assume $n_1=7$, calculate n_3 and n_5 .

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